ENERGY STAR Portfolio Manager Integration

PREPARED BY Energy Hippo implementation team

This document explains how the synthetic customer maps to ENERGY STAR Portfolio Manager (PM) so that a future integration run is mostly configuration, not modeling. The synthetic data is intentionally shaped to match the PM model that the EEMSuite EnergyStar integration already generates and consumes.

Where this is grounded

- **EEMSuite integration model** (the XSD-generated C# the integration uses): `energyhippo/EEMSuite/src/Integrations/EEMSuite.Integrations.EnergyStar/Models/Generated` — `PropertyType`, `PrimaryFunctionType`, `GrossFloorAreaType`, `OccupancyType`, and the use-detail value types.
- **EEMSuite mapping code**: `energyhippo/EEMSuite/src/Tasks/EEMSuite.Tasks.EnergyStar/Mapping/PropertyMapping.cs` — maps an EEM facility row to a PM `PropertyType`.
- **ENERGY STAR public reference**: "U.S. Property Types, Definitions, and Use Details" (Oct 2024), and the PM web services property API. Links in Sources.

The PM property model

A PM **Property** is one benchmarked building or campus. It carries:

PM field	Source	Notes
`name`	building/site display name	1–80 chars
`primaryFunction`	`PrimaryFunctionType` enum	Must be an exact enum value or it cannot be parsed
`grossFloorArea`	`GrossFloorAreaType`	`value` + `units` (`Square Feet` / `Square Meters`)
`yearBuilt`	int	
`occupancyPercentage`	`OccupancyType`	Discrete 5% increments (0, 5, ..., 100)
`isInstitutionalProperty`	bool	True for campus buildings
`address`	`AddressType`	

A Property then has one or more **Property Uses** (e.g. Office, Library, Multifamily Housing). Each Property Use carries **Use Details** — the characteristics PM uses for metrics. A single building can hold several uses (e.g. an office building with a classroom wing).

Use-detail wrapper

Every use detail derives from `UseAttributeBase`, so each value can carry:

- `value` (the measurement), and `units` where applicable.
- `currentAsOf` (date the value became current).
- `temporary` (flag for provisional values).
- `default` (read-only; PM marks values it defaulted).

Value carriers in the generated model: `UseIntegerType` (int ≥ 0), `UseDecimalType` (decimal ≥ 0), `UseStringType`, `UseYesNoType` (Yes/No), `GrossFloorAreaType`, plus typed enums such as `ResidentPopulationType`, `OnsiteLaundryType`, and `AnnualResearchExpenditureType`.

Key enum/unit values used by this dataset:

- `GrossFloorAreaUnitsType`: `Square Feet`, `Square Meters`.
- `ResidentPopulationType`: `No specific resident population`, `Dedicated Student`, `Dedicated Military`, `Dedicated Senior/Independent Living`, `Dedicated Special Accessibility Needs`, `Other dedicated housing`.
- `OccupancyType`: 0, 5, 10, … 100. `PercentHeated` / `PercentCooled`: 0, 10, … 100.
- `PoolSizeTypeValue`: `Recreational (20 yards x 15 yards)`, `Short Course (25 yards x 20 yards)`, `Olympic (50 meters x 25 meters)`. `PoolTypeValue`: `Indoor`, `Outdoor`. The Recreation and Aquatics Center pool is a short-course indoor pool.

Three grouping models in this dataset

PM supports more than one way to group buildings; the synthetic customer exercises three deliberately:

1. **Campus** — `Northlake Main Campus` is a `College/University` **campus parent** of 545,000 sq ft. Each campus building (Admin Hall, Science Center, Library, Student Center, Lakeview Residence Hall, Recreation and Aquatics Center, Parking Structure) is its own **child property** with its own `primaryFunction` and uses, rolling up to the campus. The Parking Structure is benchmarked as its own child property and is excluded from the campus floor area.
2. **Multifamily property** — `Cedar Row Apartments` is one PM-benchmarked `Multifamily Housing` **property** (124 units, `Dedicated Student`); Cedar Row A, Cedar Row B and Cedar Row Common House are its physical **buildings within the property**.
3. **Standalone property** — the `Central Plant` is its own `Other - Utility` property, benchmarked for the Northlake Thermal Plant company and excluded from the campus boundary.

Campus Grounds and Cedar Row Carport and Grounds are outdoor service areas and have no PM property.

This is captured in `data/scenarios/demo-university-v1.yaml` under `energyStarProperties` (13 entries), keyed by `siteId` / `buildingId` with a `role` (`campus_parent`, `child_property`, `multifamily_property`, `building_within_property`, `standalone_property`).

Synthetic building → PM mapping

Building		Property uses
Northlake Main Campus (site)	`College/University`	College/University, 545,000 sq ft with parking excluded (Enrollment, FTE Workers, Grant Dollars)
Admin Hall	`Office`	Office + `Other - Education` (classroom wing)
Science Center	`Laboratory`	Laboratory + `Other - Education`
Library	`Library`	Library
Student Center	`Food Service`	Food Service + `Other - Entertainment/Public Assembly`
Lakeview Residence Hall	`Residence Hall/Dormitory`	Residence Hall/Dormitory (150 rooms)
Recreation and Aquatics Center	`Fitness Center/Health Club/Gym`	Fitness Center/Health Club/Gym + `Swimming Pool`
Parking Structure	`Parking`	Parking
Cedar Row Apartments (site)	`Multifamily Housing`	Multifamily Housing (124 units, `Dedicated Student`) + `Other - Recreation` (Common House)
Cedar Row A / B	`Multifamily Housing`	Multifamily Housing (per-building)
Cedar Row Common House	`Other - Recreation`	Other - Recreation
Central Plant	`Other - Utility`	Other - Utility (standalone property outside the campus boundary)

Every `primaryFunction` and `function` string above is a verbatim `PrimaryFunctionType` enum value, so `ResolvePrimaryFunction` parses it instead of falling back to `Other`.

Use details by property type

Required vs optional per the ENERGY STAR reference (Oct 2024). Optional details do not affect metrics but are still accepted. For Fitness Center/Health Club/Gym, Swimming Pool, Parking, Other - Recreation and Other - Utility the required vs optional split has not been checked against the reference; those rows list only the use details the scenario supplies, by PM name where the manifest key table below defines one and by manifest key otherwise.

`Scored` is a property-type rule used throughout the packet: in the US a 1–100 ENERGY STAR score exists for Office, Residence Hall/Dormitory and Multifamily Housing. Library is score-eligible only in Canada; every other type in this dataset is metrics only. Per energystar.gov, a property can receive a score only when more than 50 percent of its gross floor area (excluding parking) is an eligible type and the combined floor area of uses without a score does not exceed 25 percent. Lakeview Residence Hall and Cedar Row Apartments are the properties expected to receive a score. Admin Hall exercises the second test: it is Office-primary, but its 25,000 sq ft `Other - Education` classroom wing is 29 percent of its floor area, so it gets metrics only.

Property type	Scored	Required use details	Optional use details
College/University	No	Gross Floor Area	Weekly Operating Hours, Enrollment, Number of FTE Workers, Number of Computers, Grant Dollars
Office	Yes	Gross Floor Area (≥1,000), Weekly Operating Hours (≥30), Number of Workers on Main Shift (≥1), Number of Computers (≥1), Percent That Can Be Cooled, Occupancy (≥55% to certify)	Percent That Can Be Heated
Laboratory	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Library	No (Canada only)	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers, Percent That Can Be Heated, Percent That Can Be Cooled
Food Service	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Multifamily Housing	Yes	Gross Floor Area, Total Residential Living Units (≥20), Low-/Mid-/High-rise unit counts, Number of Bedrooms, Occupancy (≥80% to certify)	Resident Population Type, Government Subsidized Housing, Laundry Hookups (in-unit / common), Percent Heated, Percent Cooled
Residence Hall/Dormitory	Yes	Gross Floor Area (≥5,000), Number of Rooms (≥5), Percent Heated, Percent Cooled	Computer Lab, Dining Hall
Fitness Center/Health Club/Gym	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers, Percent That Can Be Cooled, Percent That Can Be Heated	Not checked
Swimming Pool	No	Not checked. Scenario supplies `sizeOfPool`, `monthsInUse`, `indoorOrOutdoorPool`	Not checked
Parking	No	Not checked. Scenario supplies `completelyEnclosedParkingGarageFootage`, `partiallyEnclosedParkingGarageFootage`, `openParkingLotFootage`, `supplementalHeating`	Not checked
Other - Education	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Other - Entertainment/Public Assembly	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Other - Recreation	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift	Not checked

Property type	Scored	Required use details	Optional use details
Other - Utility	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift	Not checked

`Residence Hall/Dormitory` is assigned to Lakeview Residence Hall (150 rooms, a computer lab and no dining hall), which exercises the Number of Rooms / Computer Lab / Dining Hall details. Cedar Row stays `Multifamily Housing` with `Dedicated Student` population because it is an apartment complex, not a dorm.

Manifest field names → PM use-detail names

The `useDetails` keys in the manifest map to PM use details as follows:

Manifest key	PM use detail
`weeklyOperatingHours`	Weekly Operating Hours
`numberOfWorkersOnMainShift`	Number of Workers on Main Shift
`numberOfComputers`	Number of Computers
`percentThatCanBeCooled` / `percentThatCanBeHeated`	Percent That Can Be Cooled / Heated
`enrollment`	Enrollment
`numberOfFTEWorkers`	Number of Full Time Equivalent (FTE) Workers
`grantDollars`	Grant Dollars
`totalNumberOfResidentialLivingUnits`	Total Number of Residential Living Units
`residentialLivingUnitsLowRise` / `MidRise` / `HighRise`	Residential Living Units in a Low-/Mid-/High-rise Setting
`numberOfBedrooms`	Number of Bedrooms
`residentPopulationType`	Resident Population Type
`governmentSubsidizedHousing`	Government Subsidized Housing
`numberOfLaundryHookupsInAllUnits` / `InCommonAreas`	Number of Laundry Hookups in All Units / Common Area(s)
`numberOfRooms`	Number of Rooms
`computerLab` / `diningHall`	Computer Lab / Dining Hall

The Swimming Pool keys (`sizeOfPool`, `monthsInUse`, `indoorOrOutdoorPool`) and the Parking keys (`completelyEnclosedParkingGarageFootage`, `partiallyEnclosedParkingGarageFootage`, `openParkingLotFootage`, `supplementalHeating`) are not yet mapped to PM use-detail names here.

Integration hooks (already present in EEMSuite)

- `PropertyMapping.ResolvePrimaryFunction(string)` parses the function string by its XML enum value and falls back to `PrimaryFunctionType.Other` (a safe, unscored default) on no match — so an exact string matters.
- `PropertyMapping.ResolveOccupancy(int)` clamps 0–100 and rounds to PM's 5% scale. The synthetic `occupancyPercentage` values are already multiples of 5.

- Building mapping sets `grossFloorArea.units` by country (US → `Square Feet`).
- Property uses are serialized via `PropertyUseRequest` + `PropertyUseXmlBuilder` and read back via `PropertyUseDocument` / `PropertyUseDetailEntry`.

Not yet covered (follow-on work)

- **Meter association + consumption upload**: pushing monthly usage to PM uses `meter.xsd`, `association.xsd`, and `meterConsumption`. The synthetic meters and bills already exist (see the manifest `accounts` and provider configs); the PM side still needs meter→property association and consumption payloads.
- **Metrics read-back**: validating PM-returned metrics (Site/Source EUI, ENERGY STAR score) against expected synthetic values.
- A **negative scenario**: a building whose `primaryFunction` is an unknown string, to prove the `Other` fallback path.

Sources

- List of Portfolio Manager Property Types, Definitions, and Use Details — ENERGY STAR
- U.S. Property Types, Definitions, and Use Details (Oct 2024, PDF)
- Portfolio Manager web services — Property API